

Dependency or Phrase Structure Grammars

Stefan Müller
Humboldt Universität Berlin
Institut für deutsche Sprache und Linguistik
Dorotheenstraße 24
10117 Berlin
St.Mueller@hu-berlin.de

June 18, 2026

It is often claimed that Dependency Grammar (DG) is simpler than phrase structure grammars or even that it is “the simplest theory of grammar” (Gibson 2026). This paper shows that DGs appear simpler since many aspects of a formal theory are simply not provided. Many DGs come without a rule component. I discuss the rules that Gibson provides in a syntax textbook and show that they are formally not sound and that they mimic phrase structure-based systems in their intention. Treating semantics requires more structure either in terms of nodes or in terms of additional features. Headless constructions like Jackendoff’s NPN construction cannot be handled in DG without extensions. If projectivity is given up, as suggested by Gibson (2025), other means are needed to avoid overgeneration, again adding to complexity. Some notion of constituency is needed to ensure the necessary adjacency requirements. I discuss examples from German that suggest that V2 should be treated by some kind of “movement” rather than by simple dependency relations. It is shown that psycholinguistic arguments (e.g. Dependency Locality) cannot be used to argue against phrase structure-based approaches.

Keywords: Dependency Grammar, Syntax, theory comparison, psycholinguistics, Construction Grammar

Word count: 10883 words

1 Introduction

Gibson (2026) argues in his paper *Dependency syntax as the simplest theory of grammar* against phrase structure grammars and in particular against transformational theories.

Gibson argues for Dependency Grammar as a framework because its simplicity. If there is a given set of phenomena and properties of language that has to be described, it is easy to isolate

certain aspects and only display these. This looks rather simple in comparison to theories that take more properties into account. To take a concrete example, it is easy to draw dependency graphs for a given utterance, but a grammar has more tasks than to just provide structures for grammatical utterances. It also has to rule out ungrammatical word sequences. This is especially important since grammars are used for both parsing input and generating output. As I will show below, it is a non-trivial task to incorporate the constraints necessary for generation into (simple) Dependency Grammars.

To take a concrete example, have a look at the attested German sentence in (1):

- (1) Stets einen wachen Blick mußten Prinz Claus II. und seine Begleiter auf das
 always a woke glimps must Prinz Claus II. and his companions at the
 lange Herrschaftssymbol an der Mütze haben, damit es die sich blitzschnell
 long sovereignty.symbol at the cap have so.that it the REFL lightning-fast
 drehenden Walzen des 1300 Tonnen schweren Kolosses ja nicht
 spinning rollers of.the 1,300 metric-ton heavy colossus PART not
 verschlingen.¹
 swallow.up
 ‘Prince Claus II and his companions had to keep a close eye on the long symbol of
 sovereignty attached to his cap at all times, to ensure that it would not be swallowed up
 by the lightning-fast spinning rollers of the 1,300 metric-ton colossus.’

The sentence involves the phenomenon of multiple frontings (van de Velde 1978, Müller 2003, Bildhauer 2011), which are possible in German, although German is a V2 language (Holmberg 2015), that is, usually there is exactly one constituent before the finite verb. The adverb *stets* ‘always’ is an adjunct of *haben* ‘have’ and *einen wachen Blick* is a part of the accusative object of *haben* ‘have’. The remaining part of the object (*auf das lange Herrschaftssymbol an der Mütze* ‘on the long symbol of sovereignty attached to his cap’) is realized to the right of the finite verb. Semantically the adverb and the part of the accusative object are not connected. Both phrases depend on the verb *haben* ‘have’. If dependencies are analyzed without any further analytical devices, we get the dependency graph in Figure 1 for the relevant subpart of the sentence.²

It depends on further assumptions in the theory whether such graphs are possible at all. For example, some Dependency Grammarians assume projectivity (e.g. Hudson 2000: 23, Groß & Osborne 2009: 53). That is, crossing lines as in Figure 1 are forbidden.

A further thing that is usually missing in papers about DG (and also in papers about Government & Binding and Minimalism) is semantics. As Müller (2019) pointed out, once semantics is integrated, DG analyses look suspiciously similar to phrase structure-based analyses.

The paper is structured as follows: Section 2 discusses the fact that dependency trees do not constitute a theory. A grammar is needed to license the trees. If one looks at actual suggestions it is clear that they are not very different from phrase structure grammars. Section 3 discusses Gibson’s own proposal. Section 4 deals with claims about the alleged simplicity of DG. It shows that if semantics is integrated additional representational features or nodes are

¹Mannheimer Morgen, 03.12.2005, Ressort: Mannheim; Beim Salto Mortale herrscht Spannung pur)

²I analyzed the subject *sie* ‘they’ as a dependent of the modal verb *mussten* ‘must’ as it is done by Heringer (1996: 73). Alternatively it could be analyzed as dependent of *haben* ‘have’, which would result in one further crossing line. See Eroms (2000: Section 5.3.4) for discussion.

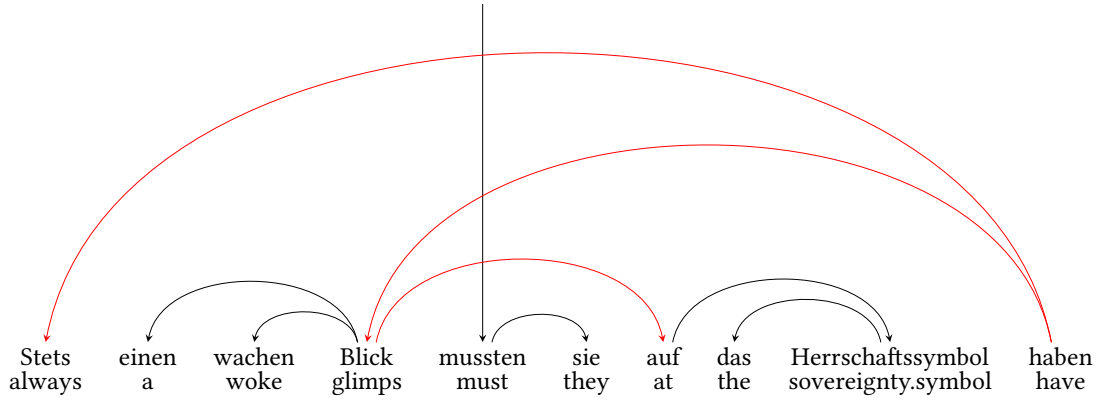


Figure 1: Analysis of an attested German example involving apparently multiple fronting together with NP split requiring crossing dependencies

needed that make the alleged advantage of DG disappear. Section 5 discusses the problem that DG has with unheaded constructions. Section 6 points out problems with discontinuity. If adjacency between heads and dependents is enforced, grammars result that are rather similar to phrase structure grammars. Section 7 discusses the V2 property of German and shows that some sort of long distance dependency is needed even for simple V2 sentences. Section 8 deals with transformations. Gibson refers to studies showing that transformations cannot be psycholinguistically real and that language acquisition problems disappear if transformations are abandoned. I point out in this section that there have been representational models in Transformational Grammar that did not have the psycholinguistic problems. Section 9 deals with computational complexity of transformational accounts and the significance of complexity results in general. Section 10 discusses Gibson’s claims regarding dependency locality and shows that the results are compatible with extraction accounts using the GPSG slash mechanism. Section 11 concludes the paper.

2 Constituents vs. dependents and the missing grammar of Dependency Grammar

I already mentioned that drawing some trees or dependency graphs is not a linguistic theory although this is what is commonly done in theoretical work on Dependency Grammar and also in Mainstream Generative Grammar (MGG). There are some researchers, however, who do provide grammar rules (Gaifman 1965: 305, Hays 1964: 513, Baumgärtner 1970: 61, Heringer 1996: Section 4.1, Gibson 2025: Chapter 3). For instance Baumgärtner suggests the general rule format in (2):

$$(2) \quad \chi \rightarrow \varphi_1 \dots \varphi_i * \varphi_{i+2} \dots \varphi_n, \text{ where } 0 < i \leq n$$

The asterisk in (2) corresponds to the word of category χ . In the example in Figure 2, χ would be V, the position of the ‘*’ would be taken by *read*, and φ_1 and φ_3 would be N. Together with the

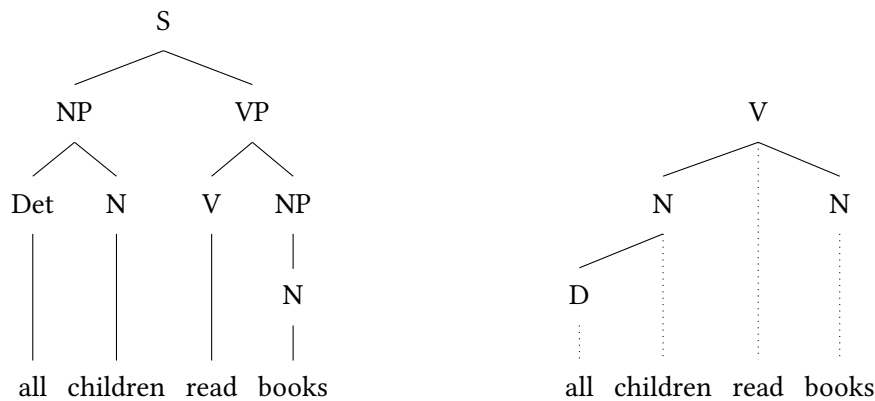


Figure 2: Phrase structure vs. Dependency Grammar analyses

rule in (3b) for the determiner-noun combination, the rule in (3a) would license the dependency tree in Figure 2.^{3,4}

- (3) a. $V \rightarrow N * N$
 b. $N \rightarrow D *$

Now compare these rules to phrase structure rules:

- (4) a. $VP \rightarrow NP V NP$
 b. $NP \rightarrow D N$

They are completely parallel. Phrase structure grammars usually do not assume completely flat representations for languages like English. They assume that the verb is combined with the objects to form a VP, which is then combined with the subject to form a complete sentence. The advantage of this analysis is that it is easy to account for adjuncts between the verb and the subject. The respective rules can be formulated recursively as in (5), which is not possible in DGs.

- (5) a. $VP \rightarrow Adv VP$
 b. $VP \rightarrow VP Adv$

If adjuncts are to be covered in rules like (3) one would need regular expressions to allow for arbitrarily many adjuncts. Huddleston & Pullum (2002: 780) provide the following examples for multiple occurrences of adjuncts.

- (6) a. For this reason, as soon as the meeting was over, he called his solicitor.
 b. He probably deeply regretted having agreed to take part.
 c. She left immediately in order to catch the early train.

³The dependency tree is displayed in a format that is common among German Dependency Grammarians (e.g., Eroms 2000). I use it in the following to make the comparison simpler.

⁴Gibson's own rules are discussed in Section 3 below.

The following data from Müller (2023c: 67) shows that there can be arbitrarily many adjuncts in NPs. One can add arbitrarily many adjectives and make the set of referents smaller.

- (7) A: All grey squirrels are big
 B: No, I saw a small grey squirrel
 A: Yes, but small grey squirrels are ill.
 B: No, I saw a healthy small grey squirrel.

There is an interesting way to generalize over the rules in (3) and (4): If dependents are represented in lists as was suggested by Bouma, Malouf & Sag (2001), one rule is sufficient:

- (8) $H \rightarrow [1], H[\text{DEPS } [1 \oplus 2], 2]$

The $\oplus$ stands for **append**, a relational constraint concatenating two lists. In (8), the **DEPS** list is split into $[1]$ and $[2]$, which can be realized before or after the head.

Bouma et al. suggest a lexical introduction of adjuncts. They have a lexical representation of arguments in a list called argument structure list (**ARG-ST**) and then use the feature **DEPS** to represent all dependents of a head. The **DEPS** list is a list that contains all the elements of the **ARG-ST** list plus a list of arbitrary length containing adjuncts (Bouma et al. 2001: 42).

The interesting thing to note here is that rules with regular expressions stand for infinitely many different rules.⁵ The lexical items of Bouma et al. (2001: 42) specify the option of arbitrarily many adjuncts. If one would apply the respective constraints, one would get an infinite lexicon. In comparison, phrase structure grammars have recursion in their rules. For example, an adjective is combined with an $\bar{N}$ and the result is an $\bar{N}$. This can be repeated as often as it is required.

- (9) $\bar{N} \rightarrow \text{Adj } \bar{N}$

If proponents of phrase structure grammar would want to work like Dependency Grammarians, they could do this and assume rules like (8), but apart from the issue with recursive structures, there are various reasons to set up grammars differently. Before I go into details, I want to point out one more thing: In order to develop a grammar like the one in (8), one has to identify the parts of a sentence or group of words. Again, the task is the same for phrase structure grammars and for dependency grammars. A common mistake made by DGers when criticising constituent tests is that they ignore the exact formulation of the tests (Gibson 2026: 4, Osborne 2018). The tests say: If a sequence of words has property X, then it is (likely that the sequence is) a constituent. This means that if the sequence does not have the property X, the test does not fail. It is simply not applicable. So, for example, Gibson – referring to Osborne (2018) – notes that some PPs cannot be clefted although they are treated as constituents. The respective phrases do not “fail constituency tests” as claimed by Gibson. The test cannot be applied. But this is not a problem as long as there is other evidence for constituent status. See Müller 2023c: Section 1.3 for more discussion of constituency tests including problematic aspects. It is true that PSGs contain more nodes than most DGs, but there are reasons for assuming them, as will be discussed below. And some of the additional nodes are motivated by the constituency tests. If

⁵See also the adjunct addition metarule of GPSG (Uszkoreit 1987: 146, Weisweber & Preuss 1992). This metarule added one adjunct at a time to flat phrase structure rules. By applying it recursively infinitely many phrase structure rules are licensed.

DG does not assume these tests, they have to come up with ad hoc solutions to the phenomena, not based on tests.

Note also that some of the Dependency Grammars are not projective. That is, they allow for crossing dependency arcs (e.g. Heringer 1996: 261, Eroms 2000: Section 9.6.2, Gibson 2025: 112, 137). Again, if this is wanted, the same can be achieved in a Phrase Structure Grammar setting. There were researchers allowing for discontinuous constituents (Zwicky 1986, Reape 1994, Kathol 2001, Müller 1999, 2004b). Within HPSG these models have been abandoned since they have problematic aspects (Müller 2024a: Section 6.4). And in any case, one has to make sure that material that belongs together is realized together (Behaghel 1909: 139) and hence one has the notion of constituent back in.

Dependency information is available in phrase structure grammars: As Figure 3 shows, the information about dependencies is recoverable. It is clear that the N *children* of *all little children* would be the highest element in a Dependency Grammar. It is the head of the noun phrase, since its head information is shared in the substructure. The determiner is a dependent of *children* since it appears in its valence list, and the adjective also depends on the noun, since its MOD value is identical to (a projection of) the noun. The same works for the verbal projection: The

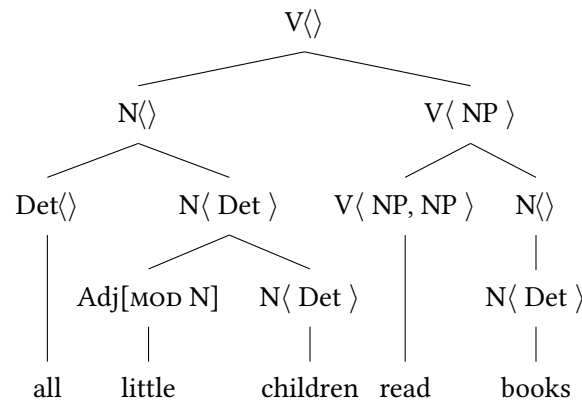


Figure 3: Dependency information is contained in (certain) phrase structure grammars

head of the complete sentence is *read*, since head information of *read* (in the figure just part of speech) is shared along the head path. The transitive verb selects two items, so it can be determined that *all little children* and *books* are dependents of *read*. The dependency information is contained in HPSG-like phrase structures and if it plays a role in psycholinguistic results, this is a good property of theories like HPSG. See Section 10 for a discussion of Dependency Locality.

3 Gibson's rules

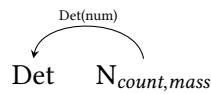
In comparison to his 2026 paper, Gibson's introduction to syntax (2025: Chapter 3) is more explicit about the rules that are assumed. Gibson suggests for every type of head a grammar rule. So there are rules for intransitive verbs, for strictly transitive verbs, for ditransitive verbs,

verbs with sentential objects and so on. There are rules combining nouns with determiners, rules for combining nouns with adjectives and for combining nouns with prepositional phrases. Gibson (2025: Section 3.14) justifies these rules by pointing to work by Goldberg (2006, 2019). He claims like proponents of phrasal Construction Grammar that the rules contribute meaning. This of course raises all the questions and causes almost all the problems of such proposals that were discussed earlier (Müller 2006, 2013, 2018, Müller & Wechsler 2014a,b).

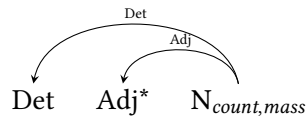
Gibson (2025: Section 3.10) explains compactness and non-redundancy. So it comes with some surprise that he suggests many many rules although only one abstract rule of the kind given in (8) would be necessary since all the relevant information about valence is represented in his lexical items already.

In addition there are some interesting facts to note about the rules Gibson suggested. Gibson shows the two dependency graphs in the Figures 4 and 5 and claims that they are derived with the rules in (10).

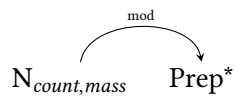
- (10) a. Rule 3, p. 58:



- b. Rule for adjectives, p. 67:



- c. Rule for prepositions modifying nouns, p. 68:



- d. Rule for prepositions modifying verbs, p. 68:

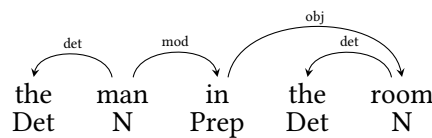
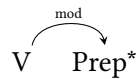


Figure 4: Dependency graphs containing adjuncts in the nominal domain according to Gibson (2025: 68)

The interesting thing about Figure 4 is that it cannot be analyzed with (10b) and (10c). The reason is that he used N twice. The problem can be seen maybe more clearly when Gibson's rules are reformulated into the format that was given above.

- (11) a. $N \rightarrow \text{Det Adj}^* *$

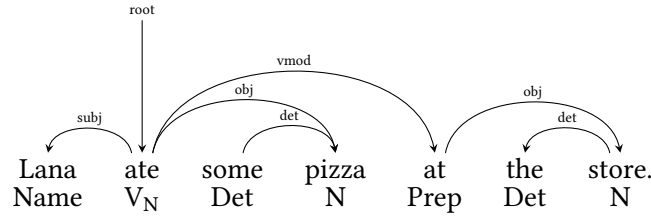


Figure 5: Dependency graphs containing adjuncts in the verbal domain according to Gibson (2025: 68)

b. $N \rightarrow * \text{Prep}^*$

The asterisk marks the position of the head, but there is no other slot for N. Either N is generated with (11a) or with (11b). It is impossible that both rules are used. Of course one can come up with formalizations that allow this kind of rule interaction. But such a formalization is not provided by Gibson. I think it is not desirable to allow for such multiple use of symbols in general. For example, if we consider languages that allow arguments at both sides of the head (e.g., Latin, Steedman & Baldrige 2006: 611), one would permit structures licensing the arguments both at the left and at the right side of the head. One could argue that semantics will take care of this, but the problem would also exist for expletive arguments, which do not play a role in semantics.

There is a further interesting aspect of rule (10c): if Gibson finds a way to allow multiple uses of head nodes, the Kleene star results into problem. First, Kleene star means zero or more occurrences of a symbol. This means that the rule in (10c) could just contain the N and no PP. So it could be applied to N arbitrarily often, making all nominal structures infinitely ambiguous. In phrase structures this would correspond to arbitrarily many applications of a unary branching rule. Gibson could go for the + instead of the Kleene star. This would require at least one PP if the rule is applied. But even this is problematic, since an NP with two PPs, that is one rule application, or as a double application of the rule adding one PP at the time. The problem worsens the more PPs are added to the noun. Note that this problem does not exist in phrase structure grammars since the result of the combination of an N with PPs can be of a different category, for example $\bar{N}$, or one can use a feature that marks the presence of PPs and forbid Ns with this feature as daughters.

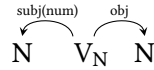
The situation with Figure 5 is even worse: it cannot be analyzed at all with the rules since Gibson assumes that there is a cost associated with crossing dependencies (Gibson 2025: 112, 137) and that projectivity is the norm since such dependencies are rare crosslinguistically (Gibson 2026: 5). The point is that the rule requires V and Prep to be adjacent (if projectivity is assumed). Gibson was probably influenced by the standard analysis in phrase structure-based systems: the verb combines with its objects to form a VP and then the VP is (recursively) modified by as many adjuncts as there may be. See (5) above. But there is no VP in DG, a fact Dependency Grammarians are proud of.

Apart from this, there is of course the problem that the verb is used twice: once in the rule for transitive verbs and once in the PP attachment rule. The way to fix this is to combine all rules that have been suggested by Gibson for combining verbs with their arguments with additional information about adjuncts. Adjuncts can appear at the beginning of the sentence, after the

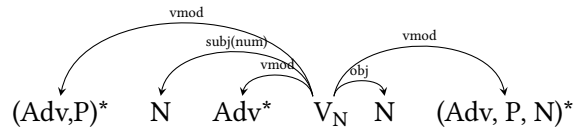
subject and clause finally. Some are also allowed between the objects.

So, Gibson would need a fusion of his (10d) with the rule for transitive verbs he provided on p. 62:

- (12) a. Rule 2, Combining a transitive verb with its arguments, p. 62:



- b. Generalized rule for transitive verbs plus adjuncts:



This means that he has to write up all the possible patterns of English sentences except the long-distance dependencies, which he analyzes via the SLASH mechanism of GPSG and HPSG (Gazdar 1981, Pollard & Sag 1987: Section 3.4). The fact that phrasal approaches have to write down all possible combinations was pointed out in (Müller 2006: 859) already and is a clear disadvantage of phrasal analyses.

Of course Gibson could assume an adjunct introduction metarule (Uszkoreit 1987: 146, Weisweber & Preuss 1992) that applies to basic argument structure constructions like (12a) and adds adjuncts. But this would add to the complexity of the used machinery and subtract from the alleged simplicity of DG.

Note that the flat rule in (12b) with an undetermined number of daughters has to be combined with a semantics, which is possible (Kasper 1994) but not trivial. This is the topic of Section 4.

Discussing semantics, Gibson (2025: 170) states that “we can think of dependency grammar as an abstraction over categorial grammar, without the semantics of the phrases”. This is not quite true, since it is impossible to please Adele Goldberg and Mark Steedman at the same time. Steedman (2000) uses very general abstract rules for the combination of lexical material. Gibson argued for very specific phrasal schemata, one schema for every possible argument structure. See also Tomasello 2003: 98–99 for explicit statements against abstract rules.

A further problem for accounts like Gibson’s that was also pointed out by Müller (2023c: Section 5.5.2) is partial verb phrase fronting. German allows for the fronting of verbs with any subset of their complements.

- (13) a. [Erzählen] wird er seiner Tochter ein Märchen müssen.
tell will he.NOM his.DAT daughter a.ACC fairy.tale must
‘He will have to tell his daughter a fairy tale.’
b. [Ein Märchen erzählen] wird er seiner Tochter müssen.
a.ACC fairy.tale tell will he.NOM his.DAT daughter must
c. [Seiner Tochter erzählen] wird er das Märchen müssen.
his.DAT daughter tell will he.NOM the.ACC fairy.tale must
d. [Seiner Tochter ein Märchen erzählen] wird er müssen.
his.DAT daughter a.ACC fairy.tale tell will he.NOM must

These examples break all phrasal approaches since ditransitive verbs have to be combined with one of their objects to the exclusion of the other one in (13b,c), unless one wants to assume non-projective structures (see Section 6). The respective other object is realized elsewhere in the clause. This is against the very nature of phrasal approaches that claim that certain aspects of the meaning are only available when a verb enters a certain corresponding phrasal construction. For (13b) this would be the phrasal construction for verbs taking a dative object and for (13c) it would be the phrasal construction for verbs taking an accusative object, but *erzählen* ‘to tell’ is a ditransitive verb in both cases. Nerbonne (1986) and Johnson (1986) solved this problem by incorporating Categorical Grammar-like treatments of valence information into GPSG. Shortly after this GPSG was given up and all researchers who worked in this framework switched to the Categorical Grammar-like framework HPSG (Pollard & Sag 1987). See Müller 2013: Section 2.3 for a comparison of Categorical Grammar, Minimalist Grammars and HPSG.

Another problem with Gibson’s rules is the analysis of coordination. Gibson (2025: Section 3.17) assumes that the coordination is the head of the dependency. So *and* with part of speech Coord ist the head of *the man and the woman*. This is problematic since the governing verb does not select for a coordination but for an N (Borsley 2005, Müller 2023b: Section 4.10.2.5). Gibson writes “the Coord category matches any syntactic head; and all conjuncts (sometimes called “coordinands”) agree with this type, including argument structures to the left and right that are being coordinated”. This is not worked out. Maybe he has in mind an analysis like the Categorical Grammar analysis in which the conjunction is a functor taking two similar arguments to project something of the same category: $(X \backslash X) / X$. But the text is not explicit about this and in any case, working this out would add to complexity.

4 Simplicity: Counting nodes and semantics

Gibson (2026) states at several places of his paper, that DG is simple. See also Osborne 2018: 6. It is not. Proponents of DG just use a subpart of what would be a complex theory if it were fully worked out.

Müller (2019: 59–60) pointed out that phrase structure is useful for determining the semantics of the unit of a head and its dependents. If the head is assigned a meaning that is usually assumed for lexical items and if there is no extra node for a phrase or constituent, then there is no place to state what the meaning of the head and its dependents is. So, while in phrase structure grammars, the meaning of *children* is something like $\lambda y. child(y)$, in DGs, it must be the meaning of the whole nominal unit (Hudson 2003: 391–392).

A way to account for this is to use a special feature for lexical meaning and one for the meaning of a head with all dependents. But this is basically shifting complexity into nodes.

Hudson develops an explicit model including semantics. He states that the representation in Figure 7 is the worked out version of the usual dependency graphs. The abbreviated form is given in Figure 8.

Now, as Müller (2019) points out, the Figure 7 with category symbols filled in as in Figure 9 suspiciously looks like an $\bar{X}$ tree. There is no advantage for DG in terms of node numbers/node complexity as soon as one looks at fully worked out theories including semantics.

Before we go on to the next section, I want to point the reader to Tesnière 2015: 150, Stemma 149.

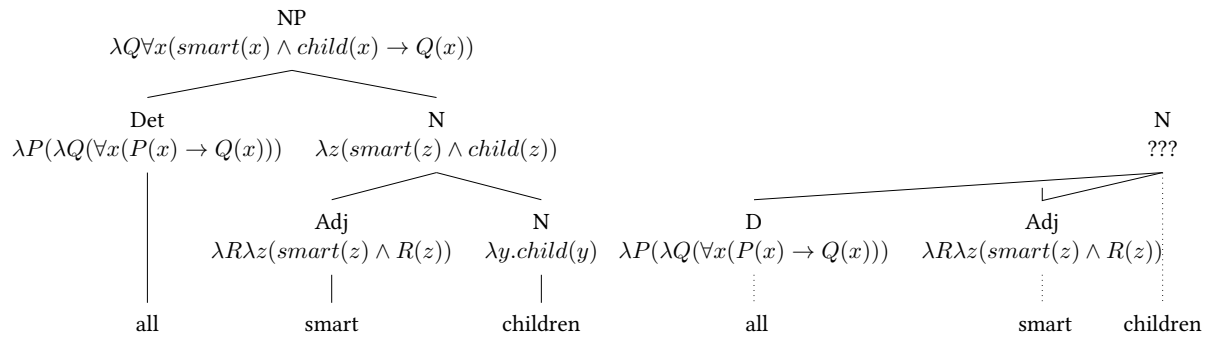


Figure 6: Nodes are used for meaning representation

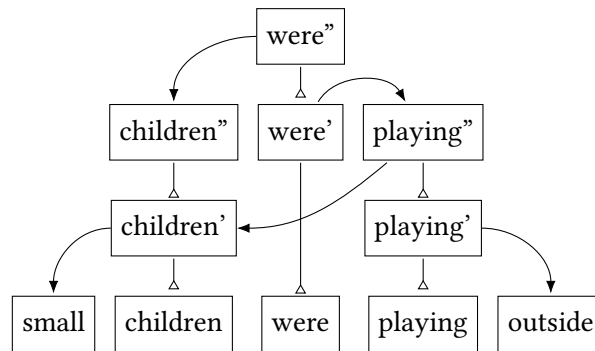


Figure 7: Analysis of *Small children were playing outside*. according to Hudson (2018: 105)

Tesnière discussed the phrase in (14) and suggested two different ways of combining the adjunct with the head for semantic reasons.

(14) red cars that you saw yesterday

So he argued for a more articulate representation (polygraphs; Kahane & Osborne 2015: lix) and not just simple dependencies and flat representations. Figure 10 shows Tesnière's representation of scope information and the corresponding phrase structure grammar.

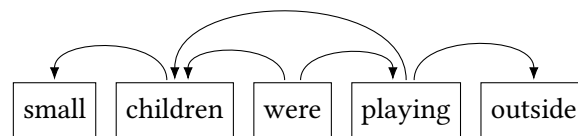


Figure 8: Abbreviated analysis of *Small children were playing outside*. according to Hudson (2018: 105)

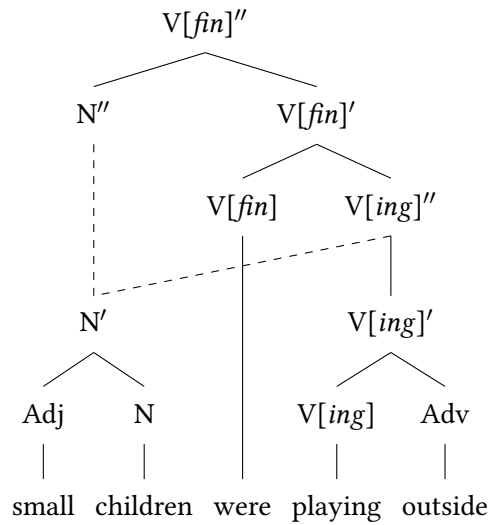


Figure 9: Analysis of *Small children are playing outside*. with category symbols

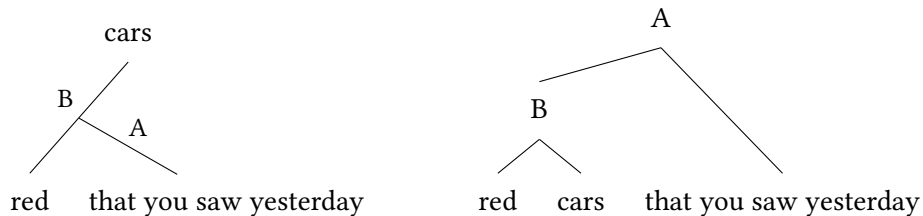


Figure 10: Tesnière's way of representing scope and the comparison with phrase structure-based analyses by Kahane & Osborne (2015: lix)

5 Dependency vs. constituency and the question of headedness

There is a clear case that decides the question of dependency vs. constituency: the question of unheaded structures. In many frameworks it is assumed that structures are headed or that there is a more prominent part that selects the other part or parts. This is true for Dependency Grammar, Categorical Grammar (Steedman 2000), and Minimalism (Chomsky 1995). This means that they do not have accounts for unheaded structures. An example is the NPN construction (Matsuyama 2004, Jackendoff 2008):

(15) Student after student after student left the room.

One property of this construction is that there can be arbitrarily repetitions of the PN part and the repetition just means 'many' without the individual PN combinations adding anything to the meaning. The nouns appear without determiner and have to have the same form of the initial noun. Unless one assumes that all parts of the construction depend on an empty N head it is not clear how these constructions can be analyzed under DG assumptions.

The first analysis of this construction in DG seems to be by Hudson (2021), but Hudson works in Word Grammar and the expressivity of this framework includes more than the head-dependent configurations that are known from DG. Hudson uses arcs to create circles. His analysis is shown in Figure 11. The lower arc from the second 1 goes back to noun_{nnp} , which accounts for the possibility to have arbitrarily many PN repetitions. As should be clear from the

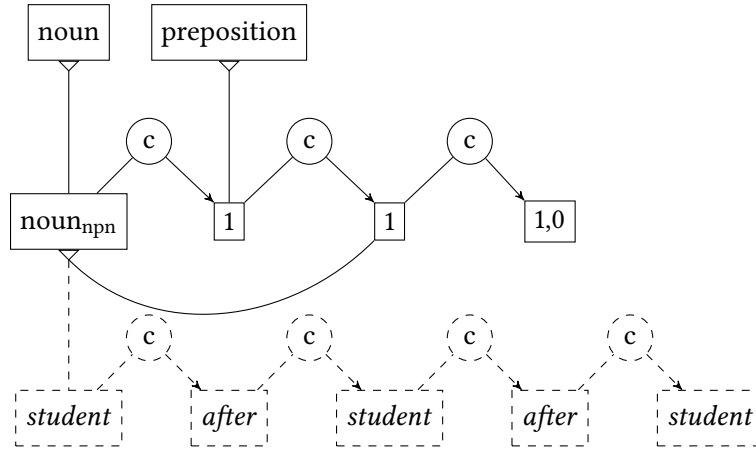


Figure 11: The NPN construction in Word Grammar according to Hudson (2021: 1477)

figure, this requires more expressivity than usual DG formalisms.

Covering the NPN construction for phrase structure-based approaches is relatively easy: There is a certain configuration with a certain meaning and a certain distribution. The material in (15) is categorized as an NP with the meaning that results from its parts (the nouns and the preposition in a certain configuration). See Bargmann (2015) for an analysis and Müller 2024b: Section 4.1 for discussion of earlier proposals in other frameworks.

For further problems for theories that are entirely head-based like Minimalism, Categorical Grammar, and Dependency Grammar see Jacobs 2008 and Müller (2023c: Section 21.10). Gibson suggested a fusion of Dependency Grammar and Construction Grammar, but without empty heads and/or other extensions, it will not be possible to capture all phenomena that are discussed in the Construction Grammar literature.

6 The intricacies of syntax: Projectivity

The success of DG in computational linguistics and in corpus linguistics is probably due to the fact that dependency annotation is simpler as phrase structure annotations where ambiguities are reflected in phrase structure. But as I hope to make clear in this paper, dependency is not enough. One aspect of competence grammars is that they can be used both ways: they can be used for parsing and for generation. Corpus annotation is just the parsing part. If discontinuities like those in Figure ?? and Figure 12 are allowed, one has to make sure that generation does not result in word salad. As Müller (2023c: 410) pointed out, non-projective DGs (Heringer 1996: 261, Eroms 2000: Section 9.6.2, Gibson 2025: 112, 137) would allow for the word salad in (16b):

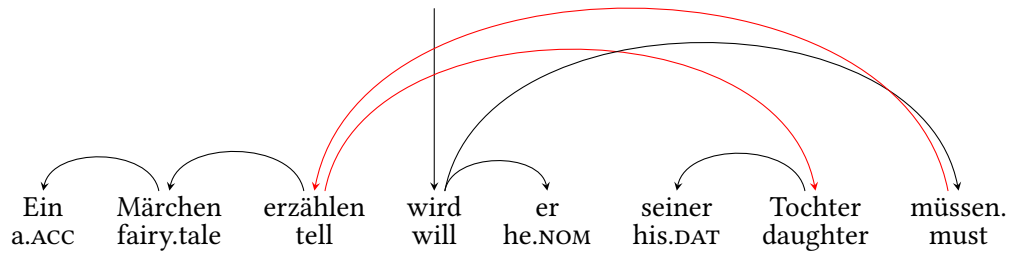


Figure 12: Analysis of (13b) requiring crossing dependencies

- (16) a. Deshalb klärt, dass Peter kommt, ob Klaus spielt.
 therefore resolves that Peter comes whether Klaus plays
 ‘That Peter comes does not resolve the question of whether Klaus comes.’
 b. *Deshalb klärt dass ob Peter Klaus kommt spielt.
 therefore resolves that whether Peter Klaus comes plays

The words from the sentences *dass Peter kommt* and *ob Klaus spielt* are all in the correct order as far as the subclauses are concerned, but they are interspersed. So, there has to be a statement somewhere in the grammar saying that both *dass Peter kommt* and *ob Klaus spielt* form continuous units.

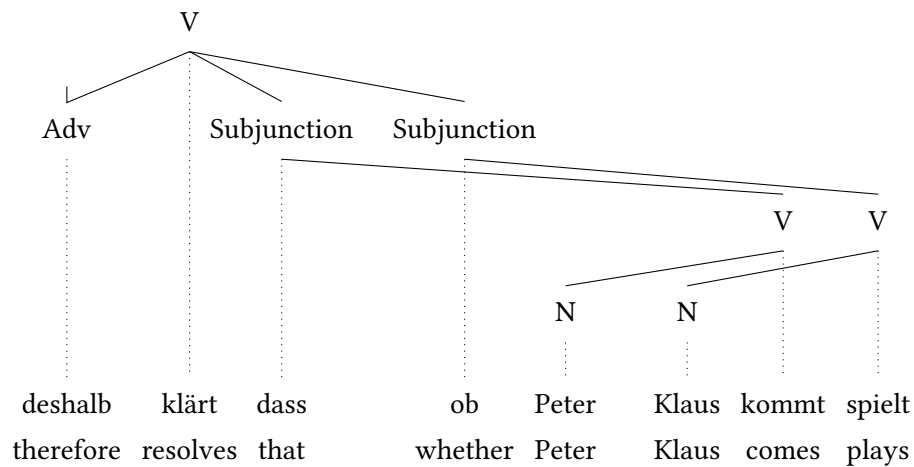


Figure 13: The dependency graph of the word salad *Deshalb klärt dass ob Peter Klaus kommt spielt*. ‘Therefore resolves that whether Peter Klaus comes plays’ which is admitted by non-projective Dependency Grammars that do not restrict discontinuity

Gibson (2026: 10) correctly points out that earlier claims about islands were wrong, but there are constraints on discontinuities. For example, extraposition is clause-bound (the Right Roof Constraint by Ross 1967: Section 5.1.2). This means that we cannot randomly place extraposed

clauses anywhere. Extraposition clearly is a nonlocal dependency (Müller 2004a, Strunk & Snider 2013), but the placement of the extraposed material is constrained.

- (17) Karl hat mir [eine Kopie [einer Fälschung [des Bildes [einer Frau _i]]]] gegeben, [die schon lange tot ist]_i.
 ‘Karl gave me a copy of a forgery of the picture of a woman who has been dead for a long time.’

It is easy to state the constraints in phrase structure grammars, since there are nodes through which the respective dependencies may not pass. If (17) is modeled with a non-projective DG it is unclear how the details can be taken care of. Kathol & Pollard (1995) develop an account for extraposition in linearization-based HPSG involving discontinuous constituents, but this involves a huge formal machinery, so any advantage in simplicity will be used up. See Groß & Osborne 2009: 84 for a formulation of the Right Roof Constraint in prose. To implement this, one would need relational constraints finding the governor of an extraposed element and checking on the way whether there are some intervening verbs.

7 V2 and “movement”

German is regarded as a V2 language. The data in (18), which is taken from Müller 2023c: Section 11.7.1 suggests that fronted material differs from material in non-fronted position:

- (18) a. Oft liest er das Buch nicht.
 often reads he the book not
 ‘It is often that he does not read the book.’ or ‘It is not the case that he reads the book often.’
 b. dass er das Buch nicht oft liest
 that he the book not often reads
 ‘It is not the case that he reads the book often.’
 c. dass er das Buch oft nicht liest
 that he the book often not reads
 ‘It is often that he does not read the book.’

(18a) has two readings, while both (18b) and (18c) are not ambiguous in the same way. Since the relative order of *oft* and *nicht* is the same in (18a) and (18c), one would not expect that (18a) has additional readings if it were just a linearization variant of (18c). The additional reading can be explained by assuming that *oft* ‘often’ is extracted with different scopings in (18a) while it is not in the other two sentences. This shows that a flat representation of a head and all its dependents as it is common of much of the DG literature on German (Heringer 1996: 261, Eroms 2000: Section 9.6.2) is not sufficient. V2 should always be handled by a special mechanism, even the apparently simple cases. Heringer and Eroms allow for non-projectivity for V2. The sentences in (18) do not require non-projectivity, but the examples show that there is more going on than simple linearization in a flat structure.

Examples like (19) show that deeply embedded material can be fronted. If one assumes projectivity, some other mechanism has to be used for “movement”.

- (19) a. [Um zwei Millionen Mark]_i soll er versucht haben, [eine
around two million Deutsche.Marks should he tried have an
Versicherung _i zu betrügen].⁶
insurance.company to deceive
'He apparently tried to cheat an insurance company out of two million Deutsche Marks.'
- b. "Wer_i glaubt er, daß er _i ist?" erregte sich ein Politiker vom Nil.⁷
who believes he that he is retort REFL a politician from.the Nile
'"Who does he think he is?", a politician from the Nile exclaimed.'
- c. Wen_i glaubst du, daß ich _i gesehen habe?⁸
who believe you that I seen have
'Who do you think I saw?'
- d. [Gegen ihn]_i falle es den Republikanern hingegen schwerer,
against him fall it the Republicans however more.difficult
[[Angriffe _i] zu lancieren].⁹
attacks to launch
'It is, however, more difficult for the Republicans to launch attacks against him.'

If one allows for non-projectivity (Heringer 1996: 261, Eroms 2000: Section 9.6.2, Gibson 2025: Section 7.7), then ways have to be found to exclude non-projective structures as the one suggested in the previous section. Gibson (2025: 183) suggests that there are preferences for projective structures, but again, this is not worked out and the mathematics of setting up a respective system is not trivial.

The sentence in (20) suggests that both simple and complex sentences should be analyzed in parallel, since a simple sentence with a gap can be coordinated with a complex sentence with a gap and the filler can fill all gaps in the conjuncts.

- (20) Wen_i kennst _i du und glaubst du, dass _i jeder kennt?
who knows you and believe you that everybody knows
'Who do you know and do you believe that everybody knows?'

So the data in (18a) and the extraction cases discussed above suggest that German V2 is a unified phenomenon both for simple and complex sentences and it involves a nonlocal dependency that differs from normal local dependencies. This is acknowledged by researchers like Groß & Osborne (2009), who use a special mechanism for nonlocal dependencies. They indicate the nonlocal dependency by a little *g* in their representations. Figure 14 shows the analysis of (19c).¹⁰ The authors call the mechanism rising but do not provide any formal details. One can imagine some GPSG-like slash passing mechanism. The DG analysis would then be equivalent to traceless GPSG and HPSG proposals (Müller 2023c: 189, Bouma et al. 2001). Gibson (2025: Section 3.19) indeed assumes a slash-based analysis for fronting in English.

⁶taz, 04.05.2001, p. 20.

⁷Spiegel, 8/1999, p. 18.

⁸Scherpenisse 1986: 84.

⁹taz, 08.02.2008, p. 9.

¹⁰The same idea can already be found in Heringer's work (1996: 261).

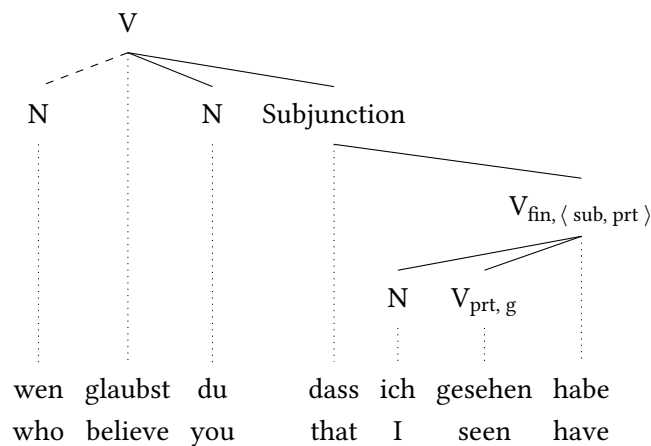


Figure 14: Projective analysis of *Wen glaubst du, dass ich gesehen habe?* ‘Who do you think I saw?’ involving rising

This brings us back to the example involving split NPs in the beginning of the paper (see Figure ??). Discussing the examples from Section 10 below, Gibson (2026) writes: “nontransformational accounts postulate direct links between words that depend on one another for meaning”. Such an approach would result in crossing dependencies as they are displayed in Figure ?? and hence in non-projectivity.

For sentences like (21), one would not get crossing edges but the V2 property of German would be violated.

- (21) Dem Saft eine kräftige Farbe geben Blutorangen.¹¹
 the juice a strong color give blood.oranges
 ‘Blood oranges give the juice a strong color.’

The dependency graph of this sentence is given in Figure 15.

The question is: What can appear before the finite verb? How many constituents? Where can they come from? A generalization about these frontings that has been confirmed in a broad study of 3,200 corpus examples (Bildhauer 2011) is that they have to depend on the same verb (Fanselow 1987: 99). Fanselow discussed the examples in (22):

- (22) a. Ich glaube dem Linguisten nicht, einen Nobelpreis gewonnen zu haben.
 I believe the linguist not a Nobel.prize won to have
 ‘I don’t believe the linguist’s claim that he won a Nobel prize.’
 b. * Dem Linguisten einen Nobelpreis glaube ich nicht gewonnen zu haben.
 the linguist a Nobel.price believe I not won to have

(22b) is out since *dem Linguisten* ‘the linguist’ depends on *glauben* ‘to believe’ and *einen Nobelpreis* ‘a nobel price’ depends on *gewonnen* ‘won’. *dem Linguisten* and *einen Nobelpreis* can be fronted alone but not together. The place in front of the finite verb is reserved for one constituent.

¹¹Bildhauer & Cook (2010) found this example in the *Deutsches Referenzkorpus* (DeReKo), hosted at Institut für Deutsche Sprache, Mannheim: <http://www.ids-mannheim.de/kl/projekte/korpora>.

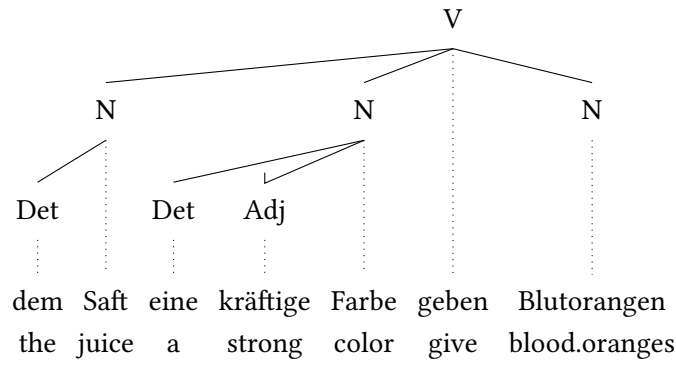


Figure 15: Dependency graph for *Dem Saft eine kräftige Farbe geben Blutorangen*. ‘Blood oranges give the juice a strong color.’

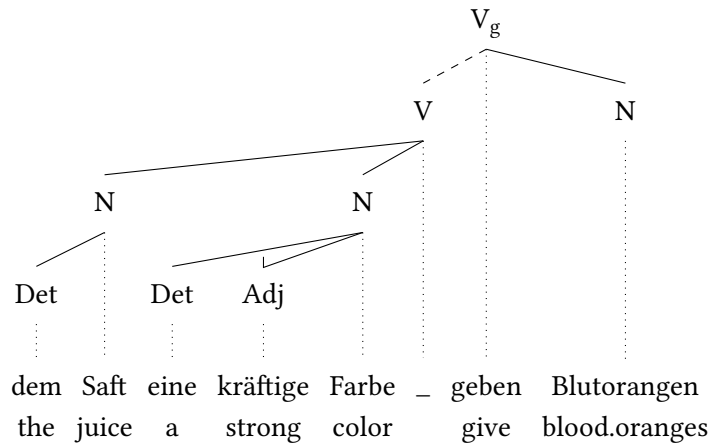


Figure 16: Dependency graph for *Dem Saft eine kräftige Farbe geben Blutorangen*. ‘Blood oranges give the juice a strong color.’ with an empty verbal head for the *Vorfeld*

How can this be modeled in DG without empty heads? See Müller (2005, 2023a) for an extensive discussion of possible analyses including multiple slash-based extractions.

Figure 16 shows a possibility for Groß & Osborne to analyze the example with an empty head. The fronted constituents would depend on an empty verbal head and this is related to a verb in the rest of the clause. Of course this begs the question what kind of empty head this is. If one assumes a verb-movement account together with formation of verbal complexes, the empty head needed for multiple frontings is just the element that is used for verb movement as well (Müller 2005: 316).

Returning to the sentence in Figure 1, this is a complex interaction of several phenomena. German allows for discontinuous NPs, which are usually analyzed via argument/adjunct attraction (De Kuthy 2002: 73–80). That is, part of an NP is raised to a higher head. In the case at hand, the PP dependent of *Blick* ‘glance’ is raised to *haben* ‘have’. As a co-dependent of *haben* ‘have’ it can be realized together with *stets* ‘always’ in fronted position.

Whatever the DG solution for the analysis of such examples will be, it is clear that it will involve much more than simply drawing lines between heads and dependents. The alledged simplicity of DG is just cherry picking.

8 Transformations

Gibson (2026) argues that the assumption of transformations is dubious. Transformations as such are known not to be psycholinguistically plausible for a long time now (Fodor, Bever & Garrett 1974: 320–328). Psycholinguists like Frazier & Clifton (1996: 27) suggested that transformations could be precompiled before actual usage, but this then means that humans work with precompiled schemata and templates that basically would correspond to what Construction Grammar suggests, but their linguistic knowledge is represented as a Transformational Grammar. This is a very implausible assumption. It begs the question why we would need the transformational grammar in the first place and do not develop theories about the “precompiled stuff”.

So, transformations mapping tree structures to other tree structures are psycholinguistically not plausible, but there have been representational variants of transformational grammar in the 1980ies and the 1990ies (Koster 1978, 1987: 235, Kolb & Thiersch 1991, Haider 1993: Section 1.4, Frey 1993: 14, Lohnstein 1993: 87–88, 177–178, Fordham & Crocker 1994: 38, Veenstra 1998: 58). Such representational variants are very similar to theories like LFG and HPSG in the versions that used a trace for extracted elements: there is a filler and somewhere else there is a corresponding trace.

With the Minimalist Programm (Chomsky 1995), Mainstream Generative Grammar took a more derivational turn again. The representational variants of Transformational Grammar are basically grammars with just one structure representing the linguistic object. Movements are indicated by coindexed empty elements. Variants of HPSG and LFG that use traces are parallel to such representational variants of MGG. The criticism against general transformations of the era before GB does not apply to those theories.

See Osborne, Putnam & Groß (2011) for a comparison of Minimalist approaches and Dependency Grammar.

9 (Transformational) Grammar and complexity results

Gibson (2026) describes transformational analyses of the passive and argues that (23a) is derived from (23b) by several transformations.

- (23) a. Lana will be kissed by Mitchell.
b. Mitchell will kiss Lana.

This is not quite true. It used to be the case that passive sentences were derived from active sentences (Chomsky 1957: 43), but in Government & Binding the passive actually had a lexical analysis (Gereon Müller, p. c. 2018). It was somehow claimed that the passive morphology blocked the subject and the participle lost the property to assign accusative. Due to the Case

Filter, the object had to move to Spec,IP to get case there (Chomsky 1981: 124). Chomsky explicitly states that the D-structure of (24a) is (24b) rather than any active sentence:

- (24) a. John was killed.
b. [NP e] was killed John.

Gibson concludes his discussion of transformations by saying “the compositional theories without such transformations are simpler than those with them [84]”. [84] refers to Peters & Ritchie 1973, which is a study of the generative capacity of general transformations as they were common in the time before Peters & Ritchie’s result. Transformations in later versions of the theory were different, they couldn’t map arbitrary trees onto other arbitrary trees and they contained traces of material that was moved (Klenk 2003: Section 3.1.4). So, these complexity results are irrelevant for current versions of Transformational Grammar.

As for current Minimalist theories, there is often a lack of formalization (Stabler 2011: 397, 399, 400) or outright inconsistency, so that complexity results cannot be obtained. An example is Chomsky’s solution to labeling (Chomsky 2008, 2013), which did not just have empirical problems but also was lacking the constraints for one possible combination of items for several years. The constraints provided in 2013 were simply inconsistent (Müller 2013: Section 2.1). There are formalized versions of Minimalist theories though and for these there are complexity results. Stabler’s Minimalist Grammars (Stabler 2001, 2011) were shown to be mildly context sensitive (Michaelis 2001). This is exactly the computational complexity that is needed, since human languages are assumed to have this complexity (Joshi 1985).

So, Gibson cannot argue against transformational theories on the basis of computational complexity. What is more, I dare say that such complexity approximations of formal systems are irrelevant for psycholinguistics anyway. They are worst-case approximations of the formalism available to the linguist, but the crucial factor is what elements of the formal language are used in an actual analysis. Such analyses are the things we should judge and the psycholinguistic predictions of which we should evaluate.

To take a concrete example, let us compare the analyses of prepositional phrases which was discussed in Minimalist textbooks (Radford 1997: 452, Hornstein et al. 2005: 124). Radford’s analysis is given in Figure 17 and the analysis in all non-MGG phrase structure grammars and Dependency Grammars is given next to it for comparison. The reader is referred to the introductory texts cited above for the explanation of the Minimalist analyses and all the empty functional categories and the involved movement operations to derive a two word phrase. See also Sternefeld (2006: 549–550) for a critical discussion.

To take another example: the formal apparatus of HPSG has Turing power, but nobody cares about this, since it is not the formal power that is available in principle but what is actually used in analyses. See Pollard 1997: 8–10 for a detailed discussion.

In any case there is enough to criticize in current Minimalist theories for psycholinguists: for example, the idea that complete Phases (Phase is a special term in Minimalist reasoning; Chomsky 2008) are build and then sent to the interfaces is incompatible with psycholinguistic findings regarding incremental processing (Marslen-Wilson 1975, Tanenhaus et al. 1996).

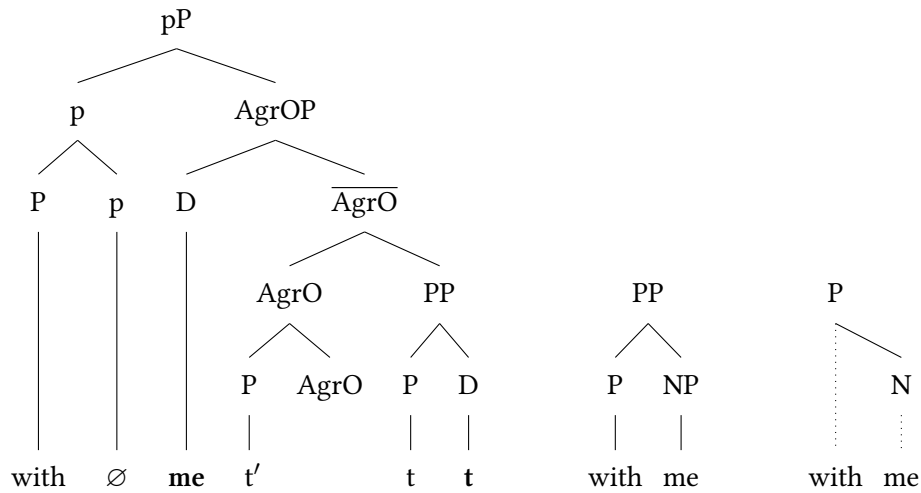


Figure 17: Minimalist PP analysis following Radford (1997: 452) compared with all other PSG and DG analyses

10 Dependency Locality

The most convincing case for DG are Gibson's examples showing that for the analysis of relative clauses the length of dependencies between relative phrase and verb is important rather than the distance between filler and gap in movement-based approaches (Pickering & Barry 1991, da Cunha & Gibson 2025). This is illustrated in Figure 18.¹²

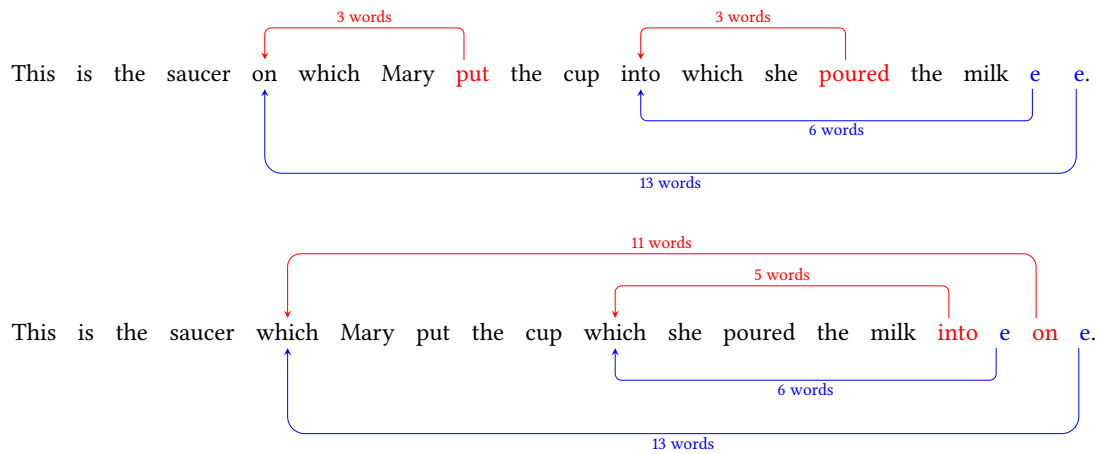


Figure 18: Distance between head and dependent vs. filler and gap

Transformational accounts (blue) predict that the processing costs for both examples in

¹²Da Cunha & Gibson (2025: 14) and Gibson (2026: 7) state that the distance between the first *which* and the trace in the second figure is 12 words, but it is 13 words as in the first figure.

Figure 18 are similar, while dependency-based accounts (red) predict that the upper sentence is more easy to process since the distance between dependent element and head is much shorter: 3 vs. 11 words.

The interesting thing now is that there are traceless approaches to nonlocal dependencies as the one by Bouma et al. (2001) in HPSG and by Sag (2010: 508) in the HPSG variant Sign-Based Construction Grammar. The information about the missing constituent is only present at the head. So the Dependency Grammar approach makes exactly the same predictions as the non-transformational Phrase Structure Grammar approach.

The traceless approach to extraction is not accepted by everybody in the HPSG community because of problems with scope of extracted adjuncts (Levine & Hukari 2006), but even approaches with actual traces for extracted elements can account for Gibson’s insights. The reason is that both *put* and *poured* contain PPs in their valence lists and the hearer/speaker of the utterance can identify the extracted element with the requirement in the verb.¹³ Note also that the information that the PPs are extracted is also represented at the verbs, since the traces have a SLASH value that is identified with the SLASH value of the PP in the valence list of the verbs. So the valence lists of the verbs look the same independent of whether empty elements are assumed or not.

This is shown in Figure 19: the verb *put* selects for two NPs and a PP. Since the PP is realized as a trace, its category is PP/PP. Since the verbs requirement is identified with the object that is actually there, overt or non-overt ([4]), the information about the SLASH value of the PP trace is available in the valence list of the verb as well. So, when the relative clause is parsed

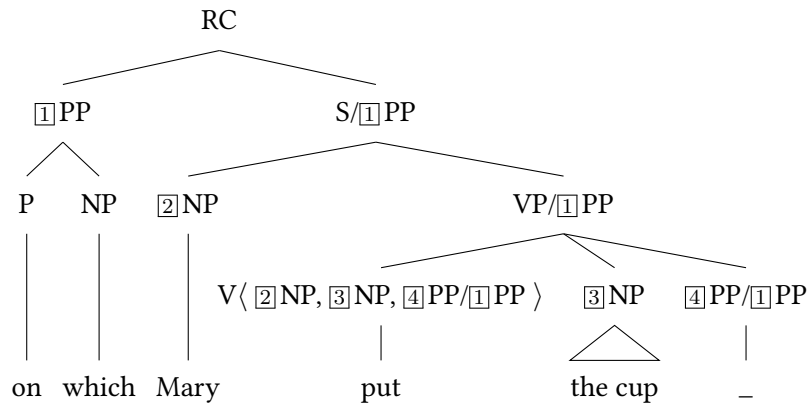


Figure 19: Information about extraction is available both at the head and at the trace.

incrementally, *on which* can be identified as the material that is needed by the verb *put*. Since the PP was first seen as a relative phrase, it is clear that it is a filler in a nonlocal dependency and it is expected that there will be a trace further to the right. To be precise: when a phrase like (25) is parsed, it is clear after the word *which* that the phrase *on which* is part of a relative clause

¹³See also Borsley’s review of Gibson 2025. Borsley (2026: 3) makes a general remark that all constituent-based frameworks with selection of sisters incorporate the notion of dependency. See also the discussion of Figure 3 above.

and a sentence missing *on which* (1) is expected. 1 is percolated down to the trace. The trace identifies the nonlocal information (1) with the local information (4). This identification is not shown in the figure, to make the “flow of information” more transparent. This is for explanatory purposes only. In reality there is no “flow”, there is just identity. 4 is shared between the trace and the requirement in the valence list of *put* and hence the fact that the PP argument is extracted is available within the lexical item for *put*.

(25) the table on which Mary put the cup

This means that it is accessible for measuring the locality of dependencies. See Borsley & Crysmann 2024 for an overview of nonlocal dependencies in HPSG and Levine & Hukari 2006 for further details on adjunct extraction.

Da Cunha & Gibson (2025: 28) refer to earlier work discussing a trace-based analysis using the principle of First-Resort Gap Positioning, but I showed another rather direct way that is possible for frameworks that register extraction at the head.

So, it seems to be the case that dependency-based and slash-based analyses of extraction are empirically indistinguishable (Sag & Fodor 1995: 376). Of course, analyses without empty elements are preferable over those with empty elements for reasons of parsimony and it is often possible to convert grammars with empty elements into grammars without any (Müller 2004c: Section 7.4), but there are situations in which they seem unavoidable or preferable (Müller 2005, 2023a, Borsley 2013: Section 4.2).

11 Conclusion

I showed that dependency relations are important but they are not sufficient to account for grammatical facts. Constraints on adjacency and projectivity are needed. It follows that claims about the alleged simplicity of Dependency Grammars in comparison to other frameworks are not warranted. Gibson was right in pointing out the psycholinguistical implausibility of Transformational Grammars but representational models of Transformational Grammars do not need literal transformations but merely empty elements and nonlocal dependencies.

Since dependency information is contained in some phrase structure grammars, the psycholinguistic results can be used for such models as well. This means that the simpler DG representations can be used for psycholinguistic purposes, but for worked out theories of language more is needed.

Acknowledgements

I thank Anne Abeillé, Bob Borsley, and Rui Chaves for discussion and pointers to further literature. I thank Sten Vikner, Matthew Korte and the participants of the Syntax-Semantics Colloquium at the Humboldt Universität zu Berlin for discussion. Thanks to Adam Przepiórkowski for comments on an earlier version of the paper and for discussion.

Competing interests

There are no competing interests.

References

- Bargmann, Sascha. 2015. *Syntactically flexible VP-idioms and the N-after-N Construction*. Poster presentation at the 5th General Meeting of PARSEME, Iasi, 23–24 September 2015.
- Baumgärtner, Klaus. 1970. Konstituenz und Dependenz: Zur Integration beider grammatischer Prinzipien. In Hugo Steger (ed.), *Vorschläge für eine strukturelle Grammatik des Deutschen* (Wege der Forschung 146), 52–77. Darmstadt: Wissenschaftliche Buchgesellschaft.
- Behaghel, Otto. 1909. Beziehung zwischen Umfang und Reihenfolge von Satzgliedern. *Indogermanische Forschungen* 25. 110–142. DOI: 10.1515/9783110242652.110.
- Bildhauer, Felix. 2011. Mehrfache Vorfeldbesetzung und Informationsstruktur: Eine Bestandsaufnahme. *Deutsche Sprache* 39(4). 362–379. DOI: 10.37307/j.1868-775X.2011.04.
- Bildhauer, Felix & Philippa Cook. 2010. German multiple fronting and expected topic-hood. In Stefan Müller (ed.), *Proceedings of the 17th International Conference on Head-Driven Phrase Structure Grammar, Université Paris Diderot*, 68–79. Stanford, CA: CSLI Publications. DOI: 10.21248/hpsg.2010.4.
- Borsley, Robert D. 2005. Against ConjP. *Lingua* 115(4). 461–482. DOI: 10.1016/j.lingua.2003.09.011.
- Borsley, Robert D. 2013. On the nature of Welsh unbounded dependencies. *Lingua* 133. 1–29. DOI: 10.1016/j.lingua.2013.03.005.
- Borsley, Robert D. 2026. Edward A. F. Gibson, *Syntax: A cognitive approach*. Cambridge, MA: MIT Press, 2025. pp. vii + 366. *Journal of Linguistics*. 1–5. DOI: 10.1017/S0022226726101200.
- Borsley, Robert D. & Berthold Crysmann. 2024. Unbounded dependencies. In Stefan Müller, Anne Abeillé, Robert D. Borsley & Jean-Pierre Koenig (eds.), *Head-Driven Phrase Structure Grammar: The handbook*, 2nd edn. (Empirically Oriented Theoretical Morphology and Syntax 9), 571–634. Berlin: Language Science Press. DOI: 10.5281/zenodo.13644930.
- Bouma, Gosse, Robert Malouf & Ivan A. Sag. 2001. Satisfying constraints on extraction and adjunction. *Natural Language & Linguistic Theory* 19(1). 1–65. DOI: 10.1023/A:1006473306778.
- Chomsky, Noam. 1957. *Syntactic structures* (Janua Linguarum / Series Minor 4). Berlin: Mouton de Gruyter. DOI: 10.1515/9783112316009.
- Chomsky, Noam. 1981. *Lectures on government and binding* (Studies in Generative Grammar 9). Dordrecht: Foris Publications. DOI: 10.1515/9783110884166.
- Chomsky, Noam. 1995. *The Minimalist Program* (Current Studies in Linguistics 28). Cambridge, MA: MIT Press. DOI: 10.7551/mitpress/9780262527347.001.0001.
- Chomsky, Noam. 2008. On phases. In Robert Freidin, Carlos P. Otero & Maria Luisa Zubizarreta (eds.), *Foundational issues in linguistic theory: Essays in honor of Jean-Roger Vergnaud* (Current studies in linguistics series 45), 133–166. Cambridge, MA: MIT Press. DOI: 10.7551/mitpress/9780262062787.003.0007.

- Chomsky, Noam. 2013. Problems of projection. *Lingua* 130. 33–49. DOI: 10.1016/j.lingua.2012.12.003.
- da Cunha, Yanis & Edward Gibson. 2025. Syntactic complexity phenomena are better explained without empty elements mediating long-distance dependencies. *Cognitive Science* 49(8). e70088. DOI: 10.1111/cogs.70088.
- De Kuthy, Kordula. 2002. *Discontinuous NPs in German* (Studies in Constraint-Based Lexicalism 14). Stanford, CA: CSLI Publications.
- Eroms, Hans-Werner. 2000. *Syntax der deutschen Sprache* (de Gruyter Studienbuch). Berlin: Walter de Gruyter. DOI: 10.1515/9783110808124.
- Fanselow, Gisbert. 1987. *Konfigurationsalität* (Studien zur deutschen Grammatik 29). Tübingen: Gunter Narr Verlag.
- Fodor, Jerry A., Thomas G. Bever & Merrill F. Garrett. 1974. *The psychology of language: An introduction to psycholinguistics and Generative Grammar* (McGraw-Hill series in psychology). New York, NY: McGraw-Hill Book Co.
- Fordham, Andrew & Matthew Crocker. 1994. Parsing with principles and probabilities. In Judith L. Klavans & Philip Resnik (eds.), *The balancing act: Combining symbolic and statistical approaches to language*, 37–41. Las Cruces, NM: Association for Computational Linguistics.
- Frazier, Lyn & Charles Clifton Jr. 1996. *Construal* (Language, Speech, and Communication 3). Cambridge, MA: MIT Press.
- Frey, Werner. 1993. *Syntaktische Bedingungen für die semantische Interpretation: Über Bindung, implizite Argumente und Skopus* (studia grammatica 35). Berlin: Akademie Verlag.
- Gaifman, Haim. 1965. Dependency systems and phrase=structure systems. *Information and Control* 8(3). 304–397. DOI: 10.1016/S0019-9958(65)90232-9.
- Gazdar, Gerald. 1981. Unbounded dependencies and coordinate structure. *Linguistic Inquiry* 12(2). 155–184.
- Gibson, Edward. 2026. Dependency syntax as the simplest theory of grammar. *Trends in Cognitive Sciences*. DOI: 10.1016/j.tics.2026.03.002.
- Gibson, Edward A. F. 2025. *Syntax: A cognitive approach*. Cambridge, MA: MIT Press. DOI: 10.7551/mitpress/15535.001.0001.
- Goldberg, Adele E. 2006. *Constructions at work: The nature of generalization in language* (Oxford Linguistics). Oxford: Oxford University Press. DOI: 10.1093/acprof:oso/9780199268511.001.0001.
- Goldberg, Adele E. 2019. *Explain me this: Creativity, competition, and the partial productivity of constructions*. Princeton, NJ: Princeton University Press.
- Groß, Thomas M. & Timothy Osborne. 2009. Toward a practical Dependency Grammar theory of discontinuities. *SKY Journal of Linguistics* 22. 43–90.
- Haider, Hubert. 1993. *Deutsche Syntax – generativ: Vorstudien zur Theorie einer projektiven Grammatik* (Tübinger Beiträge zur Linguistik 325). Tübingen: Gunter Narr Verlag.
- Hays, David G. 1964. Dependency Theory: A formalism and some observations. *Language* 40(4). 511–525.
- Heringer, Hans Jürgen. 1996. *Deutsche Syntax dependentiell* (Stauffenburg Linguistik). Tübingen: Stauffenburg Verlag.

- Holmberg, Anders. 2015. Verb second. In Tibor Kiss & Artemis Alexiadou (eds.), *Syntax – theory and analysis: An international handbook*, vol. 1 (Handbooks of Linguistics and Communication Science 42), 342–383. Berlin: Mouton de Gruyter. DOI: 10.1515/9783110377408.342.
- Hornstein, Norbert, Jairo Nunes & Kleantes K. Grohmann. 2005. *Understanding Minimalism* (Cambridge Textbooks in Linguistics). Cambridge, UK: Cambridge University Press. DOI: 10.1017/CB09780511840678.
- Huddleston, Rodney & Geoffrey K. Pullum (eds.). 2002. *The Cambridge grammar of the English language*. Cambridge, UK: Cambridge University Press. DOI: 10.1017/9781316423530.
- Hudson, Richard. 2000. Discontinuity. *Dependency Grammars*, TAL 41(1). 15–56.
- Hudson, Richard. 2003. Mismatches in default inheritance. In Elaine J. Francis & Laura A. Michaelis (eds.), *Mismatch: Form-function incongruity and the architecture of grammar* (CSLI Lecture Notes 163), 355–402. Stanford, CA: CSLI Publications.
- Hudson, Richard. 2018. Pied piping in cognition. *Journal of Linguistics* 54(1). 85–138. DOI: 10.1017/S0022226717000056.
- Hudson, Richard. 2021. HPSG and Dependency Grammar. In Stefan Müller, Anne Abeillé, Robert D. Borsley & Jean-Pierre Koenig (eds.), *Head-Driven Phrase Structure Grammar: The handbook* (Empirically Oriented Theoretical Morphology and Syntax 9), 1447–1495. Berlin: Language Science Press. DOI: 10.5281/zenodo.5599880.
- Jackendoff, Ray. 2008. Construction after Construction and its theoretical challenges. *Language* 84(1). 8–28.
- Jacobs, Joachim. 2008. Wozu Konstruktionen? *Linguistische Berichte* 213. 3–44.
- Johnson, Mark. 1986. A GPSG account of VP structure in German. *Linguistics* 24(5). 871–882. DOI: 10.1515/ling.1986.24.5.871.
- Joshi, Aravind K. 1985. Tree Adjoining Grammars: How much context-sensitivity is required to provide reasonable structural descriptions? In David R. Dowty, Lauri Karttunen & Arnold Zwicky (eds.), *Natural language parsing: Psychological, computational, and theoretical perspectives* (Studies in Natural Language Processing 1), 206–250. Cambridge, UK: Cambridge University Press. DOI: 10.1017/CB09780511597855.007.
- Kahane, Sylvain & Timothy Osborne. 2015. Translators’ introduction. In *Elements of structural syntax*, xxix–lxxiii. Translated by Timothy Osborne and Sylvain Kahane. Amsterdam: John Benjamins Publishing Co. DOI: 10.1075/z.185.
- Kasper, Robert T. 1994. Adjuncts in the Mittelfeld. In John Nerbonne, Klaus Netter & Carl Pollard (eds.), *German in Head-Driven Phrase Structure Grammar* (CSLI Lecture Notes 46), 39–70. Stanford, CA: CSLI Publications.
- Kathol, Andreas. 2001. Positional effects in a monostratal grammar of German. *Journal of Linguistics* 37(1). 35–66. DOI: 10.1017/S0022226701008805.
- Kathol, Andreas & Carl Pollard. 1995. Extraposition via complex domain formation. In Hans Uszkoreit (ed.), *33rd Annual Meeting of the Association for Computational Linguistics. Proceedings of the conference*, 174–180. Cambridge, MA: Association for Computational Linguistics. DOI: 10.3115/981658.981682.
- Klenk, Ursula. 2003. *Generative Syntax* (Narr Studienbücher). Tübingen: Gunter Narr Verlag.
- Kolb, Hans-Peter & Craig L. Thiersch. 1991. Levels and empty categories in a Principles and Parameters based approach to parsing. In Hubert Haider & Klaus Netter (eds.), *Representation and derivation in the theory of grammar* (Studies in Natural Language and Linguistic Theory

- 22), 251–301. Dordrecht: Kluwer Academic Publishers. DOI: 10.1007/978-94-011-3446-0_8.
- Koster, Jan. 1978. *Locality principles in syntax* (Studies in Generative Grammar 5). Dordrecht: Foris Publications. DOI: 10.1515/9783110882339.
- Koster, Jan. 1987. *Domains and dynasties: The radical autonomy of syntax* (Studies in Generative Grammar 30). Dordrecht: Foris Publications. DOI: 10.1515/9783110808520.
- Levine, Robert D. & Thomas E. Hukari. 2006. *The unity of unbounded dependency constructions* (CSLI Lecture Notes 166). Stanford, CA: CSLI Publications.
- Lohnstein, Horst. 1993. *Projektion und Linking: Ein prinzipienbasierter Parser fürs Deutsche* (Linguistische Arbeiten 287). Tübingen: Max Niemeyer Verlag. DOI: 10.1515/9783111355733.
- Marslen-Wilson, William D. 1975. Sentence perception as an interactive parallel process. *Science* 189(4198). 226–228. DOI: 10.1126/science.189.4198.226.
- Matsuyama, Tetsuya. 2004. The N after N Construction: A constructional idiom. *English Linguistics* 21(1). 55–84. DOI: 10.9793/elsj1984.21.55.
- Michaelis, Jens. 2001. *On formal properties of Minimalist Grammars*. Universität Potsdam. (Doctoral dissertation).
- Müller, Stefan. 1999. *Deutsche Syntax deklarativ: Head-Driven Phrase Structure Grammar für das Deutsche* (Linguistische Arbeiten 394). Tübingen: Max Niemeyer Verlag. DOI: 10.1515/9783110915990.
- Müller, Stefan. 2003. Mehrfache Vorfeldbesetzung. *Deutsche Sprache* 31(1). 29–62.
- Müller, Stefan. 2004a. Complex NPs, subadjacency, and extraposition. *Snippets* 8. 10–11.
- Müller, Stefan. 2004b. Continuous or discontinuous constituents? A comparison between syntactic analyses for constituent order and their processing systems. *Research on Language and Computation* 2(2). Special Issue on Linguistic Theory and Grammar Implementation, 209–257. DOI: 10.1023/B:R0LC.0000016785.49729.d7.
- Müller, Stefan. 2004c. Elliptical constructions, multiple frontings, and surface”=based syntax. In Gerhard Jäger, Paola Monachesi, Gerald Penn & Shuly Wintner (eds.), *Proceedings of Formal Grammar 2004*, Nancy, 91–109. Stanford, CA: CSLI Publications. <http://cslipublications.stanford.edu/FG/2004/> (3 February, 2021).
- Müller, Stefan. 2005. Zur Analyse der scheinbar mehrfachen Vorfeldbesetzung. *Linguistische Berichte* 203. 297–330.
- Müller, Stefan. 2006. Phrasal or lexical constructions? *Language* 82(4). 850–883. DOI: 10.1353/lan.2006.0213.
- Müller, Stefan. 2013. Unifying everything: Some remarks on Simpler Syntax, Construction Grammar, Minimalism and HPSG. *Language* 89(4). 920–950. DOI: 10.1353/lan.2013.0061.
- Müller, Stefan. 2018. *A lexicalist account of argument structure: Template-based phrasal LFG approaches and a lexical HPSG alternative* (Conceptual Foundations of Language Science 2). Berlin: Language Science Press. DOI: 10.5281/zenodo.1441351.
- Müller, Stefan. 2019. Evaluating theories: Counting nodes and the question of constituency. *Language Under Discussion* 5(1). 52–67. DOI: 10.31885/lud.5.1.226.
- Müller, Stefan. 2023a. *German clause structure: An analysis with special consideration of so-called multiple fronting* (Empirically Oriented Theoretical Morphology and Syntax). Berlin: Revis and resubmit Language Science Press.

- Müller, Stefan. 2023b. *Germanic syntax: A constraint-based view* (Textbooks in Language Sciences 12). Berlin: Language Science Press. DOI: 10.5281/zenodo.7733033.
- Müller, Stefan. 2023c. *Grammatical theory: From Transformational Grammar to constraint-based approaches*. 5th edn. (Textbooks in Language Sciences 1). Berlin: Language Science Press. DOI: 10.5281/zenodo.7628029.
- Müller, Stefan. 2024a. Constituent order. In Stefan Müller, Anne Abeillé, Robert D. Borsley & Jean-Pierre Koenig (eds.), *Head-Driven Phrase Structure Grammar: The handbook*, 2nd edn. (Empirically Oriented Theoretical Morphology and Syntax 9), 391–441. Berlin: Language Science Press. DOI: 10.5281/zenodo.13644960.
- Müller, Stefan. 2024b. HPSG and Construction Grammar. In Stefan Müller, Anne Abeillé, Robert D. Borsley & Jean-Pierre Koenig (eds.), *Head-Driven Phrase Structure Grammar: The handbook*, 2nd edn. (Empirically Oriented Theoretical Morphology and Syntax 9), 1497–1553. Berlin: Language Science Press. DOI: 10.5281/zenodo.5599882.
- Müller, Stefan, Anne Abeillé, Robert D. Borsley & Jean-Pierre Koenig (eds.). 2024. *Head-Driven Phrase Structure Grammar: The handbook*. 2nd edn. (Empirically Oriented Theoretical Morphology and Syntax 9). Berlin: Language Science Press. DOI: 10.5281/zenodo.13637708.
- Müller, Stefan & Stephen Wechsler. 2014a. Lexical approaches to argument structure. *Theoretical Linguistics* 40(1–2). 1–76. DOI: 10.1515/t1-2014-0001.
- Müller, Stefan & Stephen Wechsler. 2014b. Two sides of the same slim Boojum: Further arguments for a lexical approach to argument structure. *Theoretical Linguistics* 40(1–2). 187–224. DOI: 10.1515/t1-2014-0009.
- Nerbonne, John. 1986. ‘Phantoms’ and German fronting: Poltergeist constituents? *Linguistics* 24(5). 857–870. DOI: 10.1515/ling.1986.24.5.857.
- Nerbonne, John, Klaus Netter & Carl Pollard (eds.). 1994. *German in Head-Driven Phrase Structure Grammar* (CSLI Lecture Notes 46). Stanford, CA: CSLI Publications.
- Osborne, Timothy. 2018. Tests for constituents: What they really reveal about the nature of syntactic structure. *Language Under Discussion* 5(1). 1–41. DOI: 10.31885/lud.5.1.223.
- Osborne, Timothy, Michael Putnam & Thomas M. Groß. 2011. Bare Phrase Structure, label-less trees, and specifier-less syntax: Is Minimalism becoming a Dependency Grammar? *The Linguistic Review* 28(3). 315–364.
- Peters, P. Stanley & Robert W. Ritchie. 1973. On the generative power of Transformational Grammar. *Information Sciences* 6. 49–83. DOI: 10.1016/0020-0255(73)90027-3.
- Pickering, Martin & Guy Barry. 1991. Sentence processing without empty categories. *Language and Cognitive Processes* 6(3). 229–259. DOI: 10.1080/01690969108406944.
- Pollard, Carl. 1997. The nature of constraint-based grammar. *Linguistic Research* 15. 1–18. <http://isli.khu.ac.kr/journal/content/data/15/1.pdf> (10 February, 2021).
- Pollard, Carl & Ivan A. Sag. 1987. *Information-based syntax and semantics* (CSLI Lecture Notes 13). Stanford, CA: CSLI Publications.
- Radford, Andrew. 1997. *Syntactic theory and the structure of English: A Minimalist approach* (Cambridge Textbooks in Linguistics). Cambridge, UK: Cambridge University Press. DOI: 10.1017/CB09781139166706.
- Reape, Mike. 1994. Domain union and word order variation in German. In John Nerbonne, Klaus Netter & Carl Pollard (eds.), *German in Head-Driven Phrase Structure Grammar* (CSLI Lecture Notes 46), 151–198. Stanford, CA: CSLI Publications.

- Ross, John Robert. 1967. *Constraints on variables in syntax*. Cambridge, MA: MIT. (Doctoral dissertation). Reproduced by the Indiana University Linguistics Club and later published as *Infinite syntax!* (Language and Being 5). Norwood, NJ: Ablex Publishing Corporation, 1986.
- Sag, Ivan A. 2010. English filler-gap constructions. *Language* 86(3). 486–545. DOI: 10.1353/lan.2010.0002.
- Sag, Ivan A. & Janet Dean Fodor. 1995. Extraction without traces. In Raul Aranovich, William Byrne, Susanne Preuss & Martha Senturia (eds.), *WCCFL 13: The Proceedings of the Thirteenth West Coast Conference on Formal Linguistics*, 365–384. Stanford, CA: CSLI Publications/SLA.
- Scherpenisse, Wim. 1986. *The connection between base structure and linearization restrictions in German and Dutch* (Europäische Hochschulschriften, Reihe XXI, Linguistik 47). Frankfurt/M.: Peter Lang.
- Stabler, Edward P. 2001. Minimalist Grammars and recognition. In Christian Rohrer, Antje Ross-deutscher & Hans Kamp (eds.), *Linguistic form and its computation* (Studies in Computational Linguistics 1), 327–352. Stanford, CA: CSLI Publications.
- Stabler, Edward P. 2011. After Government and Binding Theory. In Johan van Benthem & Alice ter Meulen (eds.), *Handbook of logic and language*, 2nd edn. (Elsevier Insights), 395–414. Amsterdam: Elsevier. DOI: 10.1016/C2010-0-65666-5.
- Steedman, Mark. 2000. *The syntactic process* (Language, Speech, and Communication 24). Cambridge, MA: MIT Press.
- Steedman, Mark & Jason Baldridge. 2006. Combinatory Categorical Grammar. In Keith Brown (ed.), *The encyclopedia of language and linguistics*, 2nd edn., 610–621. Oxford: Elsevier Science Publisher B.V. (North-Holland). DOI: 10.1016/B0-08-044854-2/02028-9.
- Sternefeld, Wolfgang. 2006. *Syntax: Eine morphologisch motivierte generative Beschreibung des Deutschen* (Stauffenburg Linguistik 31). Tübingen: Stauffenburg Verlag.
- Strunk, Jan & Neal Snider. 2013. Subclausal locality constraints on relative clause extraposition. In Gert Webelhuth, Manfred Sailer & Heike Walker (eds.), *Rightward movement in a comparative perspective* (Linguistik Aktuell/Linguistics Today 200), 99–143. Amsterdam: John Benjamins Publishing Co. DOI: 10.1075/la.200.
- Tanenhaus, Michael K., Michael J. Spivey-Knowlton, Kathleen M. Eberhard & Julie C. Sedivy. 1996. Using eye movements to study spoken language comprehension: Evidence for visually mediated incremental interpretation. In Toshio Inui & James L. McClelland (eds.), *Information integration in perception and communication* (Attention and Performance XVI), 457–478. Cambridge, MA: MIT Press. DOI: 10.7551/mitpress/1479.003.0029.
- Tesnière, Lucien. 2015. *Elements of structural syntax*. Translated by Timothy Osborne and Sylvain Kahane. Amsterdam: John Benjamins Publishing Co. DOI: 10.1075/z.185.
- Tomasello, Michael. 2003. *Constructing a language: A usage-based theory of language acquisition*. Cambridge, MA: Harvard University Press.
- Uszkoreit, Hans. 1987. *Word order and constituent structure in German* (CSLI Lecture Notes 8). Stanford, CA: CSLI Publications.
- van de Velde, Marc. 1978. Zur mehrfachen Vorfeldbesetzung im Deutschen. In Maria-Elisabeth Conte, Anna Giacalone Ramat & Paolo Ramat (eds.), *Wortstellung und Bedeutung: Akten des 12. Linguistischen Kolloquiums, Pavia 1977* (Linguistische Arbeiten 61), 131–141. Tübingen: Max Niemeyer Verlag. DOI: 10.1515/9783111713083.131.

- Veenstra, Mettina Jolanda Arnoldina. 1998. *Formalizing the Minimalist Program*. Rijksuniversiteit Groningen. (Ph.D. thesis).
- Weisweber, Wilhelm & Susanne Preuss. 1992. Direct parsing with metarules. In Antonio Zampolli (ed.), *Proceedings of the 15th International Conference on Computational Linguistics (COLING '92)*, August 23–28, 1111–1115. Nantes, France: Association for Computational Linguistics.
- Zwicky, Arnold M. 1986. Concatenation and liberation. In Anne M. Farley, Peter T. Farley & Karl-Eric McCullough (eds.), *Papers from the 22nd Regional Meeting of the Chicago Linguistic Society*, 65–74. Chicago, IL: Chicago Linguistic Society.